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1).

Autonomous Vehicle

Computer Vision Overview

- It enables machines to interrupt visual information.
- Uses cameras to capture road scenes.
- Helps autonomous vehicles understand surroundings.

Image Sensing & Acquisition

- Cameras capture road images / videos.
- Lens focus light onto the image sensor.
- Sensor converts light into electrical / digital signals.

Digital Image fundamentals

- Image represented as pixels.
- Each pixel has intensity colour values.
- Resolution determines image detail.

Sampling and Quantization.

- Sampling converts continuous image into pixels.
- Quantization converts intensity into the discrete values.

Features of computer vision

- Edge detection
- Shape and texture analysis.
- Feature extraction.
- Object detection and recognition.
- Robustness to changes in lighting.

Levels of computer vision and image

- Low-level : Image enhancement filtering.
- Mid-level : feature extraction and segmentation.
- High-level : object recognition.

Computer Vision in Autonomous

- Captures - process →
- Extracts features.
- Support Extract progression.

Object Recognition.

- Detects pedestrains, vehicles, bycycles.
- Recognizes traffic signs and signals.
- Identifies road boundaries and obstacles.

Navigation & Decision Making

- Understands road & traffic conditions.
- Determines safe path and direction.
- Supports steering, breaking and acceleration.

Applications

* Autonomous cars.
* Lane detection
* Traffic sign recognition.
* Pedestrain detection

2).

Medical Imaging

Computer Vision overview

- Enables computers to interupt medical images.
- Extracts useful information from X-rays.
- Helps detect abnormalities.
- Supports diagnosis and clinical decision-making.

Image Sensing & Acquisition

- X-ray source produces X-rays.
- X-rays pass through patient's body.
- Detects captures transmitted X-rays.

Digital Image fundamentals

- X-ray image consists of pixels.
- Pixel intensity represents X-ray attenuation.
- Resolution determines anatomical detail.

Sampling and Quantization

- Sampling converts the continuous x-ray image into pixels.
- Quantization assigns discrete intensity levels.

Levels of computer Vision processing.

- Low-level: noise reduction, enhancement.
- Mid-level: segmentation.
- High-level: abnormality detection.

Medical Image analysis

- Detects lesions and abnormalities.
- Identifies changes in bone and soft tissue.
- Measures size, shape and location of lesions.

Image Processing & feature extraction

- Edge detection
- Feature extractive.
- Shape pattern.

Applications

* Fracture detection
* Lung disease detection
* Chest x-ray analysis.
* Computer-aided diagnosis.

DIP (In context)

- Capturete edge representation.
- Pixel intensity issues.

Image formation models

- Describes how 3-D objects from 2-D images
- Formates Completion merge.
- Perspective projection represents road scene.

Overall outcome

- Clear enhancial dignified x-ray image.
- Accurate and detection.